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2 **Reassessing the Role of Progenitor Cells in**  
3 **Muscle Wasting in Cancer Cachexia**  
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## ABSTRACT

Skeletal muscles contain myogenic and non-myogenic progenitor cells that proliferate and differentiate after tissue damage to restore myofiber, connective tissue, and blood vessel homeostasis. We previously showed that cancer-induced muscle wasting involves myofiber damage and impaired differentiation of myogenic progenitors, coincident with the aberrant accumulation of mesenchymal progenitor cells expressing both myogenic (*Pax7*) and non-myogenic (*Sca1*, *Pdgfra*) progenitor markers. Here, we combined lineage tracing and scRNA-seq to more deeply explore muscle resident progenitor cells during cancer cachexia. Colon-26 (C-26) carcinoma cells were injected into adult ( $\geq 12$ -weeks-old) *Pax7*-Cre<sup>ER</sup>; *Rosa26*-LSL-*tdTomato* mice, and tamoxifen was administered after tumors were established but prior to cachexia. At cachexia endpoint, scRNA-seq analysis was performed on muscle mononuclear cells. In both control and C-26 muscles, *Pax7* and *tdTomato* transcripts were restricted to myogenic progenitors, whereas *Sca1* and *Pdgfra* were restricted to non-myogenic progenitors. These results were confirmed by flow cytometry and suggest that mesenchymal progenitor cells do not commit to a myogenic fate during cancer cachexia. However, consistent with earlier findings, our transcriptomic analyses validated that myogenic progenitors from tumor-bearing mice were impaired to differentiate. When we repeated *Pax7*-lineage tracing and scRNA-seq on young mice (6-week-old) still undergoing developmental muscle growth similar results were obtained, but interestingly, by flow cytometry we detected a small population of *Sca1*<sup>+</sup>; *tdTomato*<sup>+</sup> cells, not present in adult muscles. Thus, non-myogenic progenitors might indeed be capable of adopting a myogenic fate during cancer cachexia, but this contributes to only a minor fraction of *Pax7*<sup>+</sup> cells and likely to be age dependent.

## **New and Noteworthy**

Here, we performed Pax7-lineage tracing and unbiased scRNA-seq on skeletal muscle mononuclear cells during cancer cachexia. We found myogenic progenitors from cachectic muscle have an activation-like transcriptional profile, and concurrent downregulation of terminal differentiation genes. However, contrary to previous findings, aberrant commitment of non-myogenic progenitors to a myogenic lineage did not occur in cachectic muscle. Subsequent experiments demonstrated inflammation and impaired differentiation of activated myogenic progenitors occur early in tumor progression, prior to muscle wasting.

## INTRODUCTION

Rapid loss of skeletal muscle mass and function are central features of cancer cachexia, a highly prevalent side effect of tumor progression that reduces patient quality of life, treatment options, and survival time (1). Cachexia is estimated to occur in more than half of all cancer patients, with certain cancers such as pancreatic ductal adenocarcinoma (PDAC) with incidence rates as high as 80% (1). In addition to reducing patient functional capacity and treatment efficacy, muscle wasting directly contributes to poor outcome in cancer patients due to cardiorespiratory failure (2-4). However, effective therapies for preventing muscle wasting and dysfunction are still lacking. As such, increasing our understanding of the molecular mechanisms mediating muscle degeneration is critical for improving treatment and survival of cancer patients suffering from cachexia.

Skeletal muscle tissue contains multiple progenitor cell populations that maintain tissue homeostasis throughout adulthood by enabling repair/turnover of damaged myofibers and stromal components. Myofiber repair is accomplished via the activation and differentiation of muscle stem cells (MuSCs), commonly known as satellite cells, which reside between the myofiber sarcolemma and basal lamina (5). These quiescent cells are activated by various physical and chemical changes to their niche environment, resulting in their activation and proliferation into a heterogeneous population of daughter cells that either commit to terminal myogenic differentiation and fuse into multinucleated myofibers, or maintain their stemness and undergo self-renewal to replenish the MuSC pool (6).

MuSCs and non-committed myogenic progenitors are marked by the expression of paired box 7 (Pax7), a transcription factor that is necessary for maintaining myofiber regeneration throughout adulthood due to its master control of transcription factors such as MyoD, myogenin (MyoG) and Myf5, which control myogenic lineage specification, progression, and terminal differentiation (7-9). We previously found that muscles from cancer patients and mice with cachexia exhibit damaged sarcolemma and increased myofiber permeability (10, 11).

Consistent with these markers of myofiber damage, cachectic muscles had significant greater numbers of proliferating Pax7<sup>+</sup> cells (10). An increase in Pax7<sup>+</sup> cells in muscles during cancer cachexia has been reported by others as well (12-14). However, rather than differentiating and fusing into damaged myofibers, Pax7<sup>+</sup> cells remained elevated, coinciding with impaired induction of the transcription factors MyoD and MyoG, which mediate terminal myogenic differentiation. The inability of myogenic progenitors to downregulate Pax7 prevents MyoD from inducing terminal differentiation (8, 15, 16). As such, conditional deletion of Pax7 rescued differentiation, leading to restoration of MyoD and MyoG and an increase in myoblast fusion to myofibers and myofiber size (10).

In addition to MuSCs, skeletal muscles contain several other mesenchymal progenitor cell populations that reside in the interstitial and/or perivascular spaces such as fibro-adipogenic progenitors (FAPs) (17, 18), pericytes (19, 20), PW1<sup>+</sup> interstitial cells (PICs), mesoangioblasts (21, 22), and muscle-derived cell stems (MDSCs) (23, 24). While these interstitial cells lack Pax7 expression, multiple groups have demonstrated their ability to contribute to myogenesis during postnatal muscle growth or following transplantation into injured muscles of juvenile mice (19-21, 23-25). Interestingly, in our previous study, damaged myofibers from cancer cachectic muscles also had abundant interstitial cells expressing both myogenic (Pax7) and non-myogenic progenitor cell markers (Sca1, Pdgfra, NG2) (10). This suggested that in a cancer setting, non-myogenic progenitors were able to commit to a myogenic fate, and like MuSCs, were impaired to differentiate.

To gain additional insight into the populations of non-myogenic progenitor cells that convert to a myogenic fate, we designed a strategy using Pax7-lineage tracing with scRNA-seq in a standard mouse model of cancer cachexia. Obtained results challenged our original findings, leading us to re-evaluate the role of Pax7<sup>+</sup> myogenic and non-myogenic progenitor cells in cancer cachexia.

## METHODS

### *Animal models of cancer cachexia*

Animals were housed with littermates at a constant temperature of 72°F, 12-12 light dark cycle, and ad libitum access to standard chow diet and water. All experiments were conducted using male mice (6-20 weeks-old) from either CD2F1 or C57BL/6J genetic backgrounds. For the C-26 model, vehicle (PBS) or  $1 \times 10^6$  colon 26 carcinoma cells (C-26) were subcutaneously injected into the right flank of CD2F1 mice as previously described (26, 27). Animals were monitored daily and changes in body weight were measured. Cachexia was determined at sacrifice as tumor-free body weight loss  $\geq 10\%$  initial body weight, or as indicated. For the Lewis Lung Carcinoma (LLC) model, vehicle (PBS) or  $5 \times 10^5$  Lewis lung carcinoma cells were intramuscularly injected into the right gluteus of C57BL/6J mice. For the orthotopic pancreatic cancer cachexia model,  $1 \times 10^3$  murine KPCML1 pancreatic cancer cells or an equal volume of Matrigel were injected into the head of the pancreas as we have recently described (28). Animals were euthanized by CO<sub>2</sub> asphyxiation. All experiments were approved by the Medical University of South Carolina Institutional Animal Care and Use Committee.

### *Lineage tracing of myogenic progenitors*

Labelling and lineage tracing of myogenic progenitor cells was performed using Pax7-CreER; Rosa26-LSL-tdTomato mice (Jackson Lab Strain #:017763). To activate Cre recombinase and induce tdTomato expression in Pax7-expressing cells, animals were I.P. injected with tamoxifen (Sigma T5648, 1mg/10g b.w.) dissolved in corn oil once daily for 5 consecutive days.

### *Muscle mononuclear cell isolation*

Muscles from both hindlimbs were dissected and placed in digestion buffer (serum free DMEM, 2mg/mL Collagenase II, 2 U/mL Dispase) on ice. Muscles were then manually minced

with scissors before placing on a Miltenyi dissociator for 1 h at 37°C to complete digestion. After digestion, samples were diluted with serum supplemented DMEM then sequentially filtered through 70um and 40um cell strainers. Samples were then centrifuged at 1500 rpm for 5 minutes to pellet cells and then resuspended in FACS buffer (PBS+0.5% BSA+2mM EDTA). Cell suspensions were slowly transferred into conical tubes containing horse serum and centrifuged for 10 minutes at 1000 rpm without brake. Cell pellets were resuspended in PBS+0.5% BSA and placed on ice before proceeding to scRNA-sequencing or flow cytometry/FACS analysis.

### ***scRNA-seq and data analysis***

Raw sequences were processed with Cell Ranger (v7.1.0). Cellranger mkfastq command was used to demultiplex the different samples and cellranger count command was used to generate gene-cell expression matrices. The sequences were aligned to the GRCm38 (mm10) reference genome, modified to include the tdTomato transgene. Ambient RNA contamination was inferred and removed using CellBender (v0.232) with standard parameters. Downstream analysis was performed in R with Seurat (v4.3.0.1) and customized R scripts. Data from control and C-26 samples were merged into a unique single cell object. Cells with >200 genes, >2000 UMIs, and <20% mitochondrial transcripts were retained for downstream analysis. Genes located in the mitochondrial genome were removed. Doublets were removed using scDblFinder (v1.14.0). Prior to integration, the data was normalized and scaled, regressing out variables of number of UMIs and percent of mitochondrial gene expression. The data was integrated via Seurat, using Canonical Correlation Analysis (CCA) to find anchors for integration of the 2 samples. After integration, principal component analysis (PCA) was used to cluster all cells. PCA was conducted using 30 principal components and a resolution of 0.3 for Louvain clustering and UMAP. Cluster markers were identified using the package Presto and Wilcoxon Rank Sum test.

### ***MuSC subclustering and pseudotime trajectory analysis***

MuSCs were subset from the integrated dataset and re-clustered using the first 30 principal components and a resolution of 0.5 for Louvain clustering, resulting in three distinct subpopulations. Normalized counts were used for differential expression analysis and Presto function to identify subcluster gene markers. Pseudotime trajectory analysis was performed using Monocle3 version 1.4.26. Cells were ordered along pseudotime using the quiescent population (Cluster 1) as the root state to reconstruct differentiation trajectories. To assess condition-specific dynamics, pseudotime was computed independently for each condition.

### ***Gene Set Enrichment Analysis (GSEA)***

Transcriptomes from Pax7<sup>+</sup> or Myf5<sup>+</sup> cells were filtered and the normalized mean expression matrix (excluding genes expressed in < 20% of cells) was input into the GSEA software (29) . Log2 ratio of normalized mean expression, 1,000 permutations, gene set permutation type, and weighted enrichment, were used in the analyses. Gene sets containing 15 to 500 genes were included and Hallmarks and M5 Gene Ontology databases were used. Gene sets with p-value <0.05 and FDR <0.25 were considered significant.

### ***RNA sequencing***

TdTomato<sup>+</sup> cells were isolated by FACS and sorted directly into tubes containing Trizol LS reagent (Life Technologies, CA) and stored at -80°C. Total RNA was isolated from cells according to manufacturer recommended protocol. RNA quality was assessed using an Agilent 4200 TapeStation and RINe values ranged from 8–10. Total RNA was utilized to construct cDNA libraries with the New England Biolabs NEBNext® Poly(A) mRNA Magnetic Isolation Module (Cat# 7490L) and Ultra II Directional RNA Library Prep Kit from Illumina (Cat# 7760L) as recommended by the manufacturer. Dual-indexed libraries were pooled and sequenced at



VANTAGE (Vanderbilt University Medical Center) on an Illumina NovaSeq 6000 to a depth of approximately 25 million paired-end 150 bp reads per library.

### ***Flow cytometry and Florescence Activated Cell Sorting***

Sca1 (Ly6a) and tdTomato were detected on freshly isolated muscle mononuclear cells. Mononuclear cells from control or cachectic muscles were resuspended in blocking solution (PBS+2%FBS) and then incubated with Ly6a-PE-Cyanine7 conjugated antibodies (Invitrogen, 25-5981-82) in blocking solution for 30 min on ice in the dark. For quantification of tdTomato cells, freshly isolated mononuclear cells were fixed with 10% neutral buffered formalin for 30 min on ice in the dark, then washed and resuspended in FACS buffer. tdTomato<sup>+</sup> cell counts per 500K cells were quantified by flow cytometry and normalized to total muscle tissue weight. For RNA and protein expression analysis, fluorescence activated cell-sorting (FACS) was used to capture tdTomato<sup>+</sup> and tdTomato<sup>-</sup> mononuclear cells. Cells were collected in conical tubes containing PBS+0.5% BSA on ice and then pelleted by centrifugation. Supernatants were removed and cell pellets were placed at -80°C until RNA/protein extraction.

### ***Real Time PCR***

Total RNA was extracted from FACS isolated mononuclear cells using Trizol reagent (Life Technologies) according to the manufacturer's protocol. cDNA was synthesized using M-MLV Reverse Transcriptase (Life Technologies) and real-time PCR was performed using SYBR Green (Roche) and Applied System QuantStudio3. Gene expression values were normalized to  $\beta$ -actin expression.

### ***Western blotting***

Proteins were extracted from tibialis anterior muscles or isolated mononuclear cells using RIPA buffer containing protease inhibitors (Abcam) and analyzed by SDS-PAGE as done

previously (30). Equal loading and transfer of proteins was confirmed by Ponceau S staining and Gapdh expression levels. Primary antibodies for Gapdh (Santa Cruz, 32233) and Sca1 (Invitrogen, 14-5981-82), were diluted in TBS-T 5% BSA solutions. Secondary IgG HRP conjugated antibodies (Promega-W402B, W401B) were diluted in TBS-T 2% BSA. Protein bands were detected using Immobilon Western Chemiluminescent HRP Substrate (Millipore, Cat# WBKLS0100) and a Bio-Rad Chemi Doc imager. Molecular weights were determined using Precision Plus Protein ladder (Bio-Rad, Cat#1610374).

## **Statistics**

All results are expressed as mean  $\pm$  SD. Statistical comparisons were conducted using two-tailed independent t-tests between groups of two and  $p$ -value  $<0.05$  was considered significant. Data were analyzed using R or GraphPad Prism 10 software.

## **RESULTS**

### ***scRNA-seq analysis indicates non-myogenic progenitors do not commit to a myogenic fate during cancer cachexia***

Previous findings indicated that non-myogenic progenitor cells acquire Pax7 expression and a myogenic fate during cancer cachexia, and similar to MuSCs had impaired myogenic differentiation which contributed to muscle atrophy in tumor-bearing mice (10). To identify all possible non-myogenic progenitor cell populations subject to this regulation, we combined Pax7 lineage tracing with scRNA-seq in C-26 tumor bearing mice (Fig. 1A). Tumor cells were injected into *Pax7-Cre<sup>ER</sup>; R26-LSL-tdTomato* mice and tamoxifen was subsequently administered 12-17 days-post-injection (DPI) to trace Pax7<sup>+</sup> progenitors and their progeny using tdTomato expression. At endpoint (21 DPI), when mice had developed cachexia (Fig. 1B), mononuclear cells were isolated from hindlimb muscles and scRNA-seq was performed. After filtering low-quality cells and doublets, transcriptomes from 16,427 (control) and 11,528 (C-26) cells were

integrated. Uniform Manifold Approximation and Projection (UMAP) analyses yielded 18 distinct cell clusters (Figs. 1C and D). Biomarker identification analysis and manual inspection of known cell-specific genes revealed 14 cell types including MuSCs/myogenic progenitors, endothelial, FAPs, pericytes, smooth muscle, neural glial/schwann cells, tenocytes, neutrophils, T-cells, NK-cells, macrophages, B-cells, basophils/mast cells, and mature skeletal muscle (Figs. 1D and E). As expected, cachectic muscles showed several signs of tissue damage, such as increased expression of proinflammatory cytokines/chemokines (*Ccl2*, *Cxcl1*, *Il6*) and complement pathway activation genes (*C3*, *C4b*, *C7*) in FAPs (Supplementary Figs. 1A and B) (31, 32), and elevated infiltrating pro-inflammatory monocytes/macrophages (*Ccr2*<sup>+</sup>, *Ly6c2*<sup>+</sup>, *Il1β*<sup>+</sup>) (Supplementary Figs. 1C and D) (33-35). With regards to identification of mesenchymal progenitor cell populations, *Ly6a* (*Sca1*) was expressed by FAPs and pericytes as expected, while *Pdgfra* transcripts were detected exclusively in FAPs (Fig. 1G and 1H). However, contrary to our previous findings, these mesenchymal progenitor populations lacked expression of *Pax7* and *tdTomato* in both control and cachectic muscles (Figs. 1F and H), suggesting that they do not acquire a myogenic fate during cancer cachexia.

#### ***Flow cytometry analysis confirms that non-myogenic progenitors do not adopt a myogenic fate during cancer cachexia***

Although scRNA-seq is a powerful technique for identifying distinct cell populations within tissues, it does have certain limitations such as only detecting RNA and the inability to sequence all cellular transcripts in every cell due to finite sequencing depth capacity. These limitations could prevent the detection of rare cell types that express genes at levels below the sequencing depth (36), or cells that have acquired genes via cell-cell exchange of proteins via exosomes or nanotubules (37-39). Therefore, we used flow cytometry and fluorescence-activated cell sorting (FACS) to verify our scRNA-seq results. Mononuclear cells were isolated from muscles of control or cachectic *Pax7*-tdTomato reporter mice, and cells expressing *Sca1* and/or tdTomato

were quantified by flow cytometry (Fig. 2A). As before, results showed neither control nor cachectic C-26 muscles contained mononuclear cells that co-expressed Sca1 and tdTomato (Fig. 2B). Moreover, in FACS-isolated tdTomato<sup>+</sup> and tdTomato<sup>-</sup> cell populations, *Sca1* and *Pdgfra* mRNA were detected exclusively in the tdTomato<sup>-</sup> cell fractions, while *Pax7* and *tdTomato* mRNA were detected only in tdTomato<sup>+</sup> cells (Fig. 2C). The lack of Sca1 protein expression in tdTomato<sup>+</sup> cells was also confirmed by western blotting (Fig. 2D). These findings are in line with our scRNA-seq results and suggest that skeletal muscle mesenchymal progenitors do not acquire Pax7 expression and commit to a myogenic fate during cancer cachexia as previously concluded.

### ***MuSCs are activated during cancer cachexia but exhibit impaired terminal differentiation***

Based on these revised findings, we next sought to reassess the regulation of muscle progenitor cells during cancer cachexia. Our previous findings that MuSCs were dysfunctional during cancer cachexia stemmed from electron micrographs showing myofiber sarcolemma damage, and increased sarcolemma immunoglobulin (IgG) permeability, in both mice and patients undergoing cancer cachexia (10). IgG deposition in muscle is an established consequence of myofiber damage during genetic and autoimmune-related inflammatory myopathies, and more recently has been described to occur during aging in both muscle and adipose tissue (40-42). As such, we measured IgG protein levels in skeletal muscles over time in C-26 tumor-bearing mice, as a readout for tissue damage and inflammation. Results showed noticeable IgG accumulation by 10 DPI, which increased progressively and substantially until cachexia endpoint (Figs 3A). This finding was also confirmed during LLC-tumor progression and at endpoint in an orthoptic KPCML1 mouse model of PDAC-induced cachexia (Fig. 3B and C) (28), demonstrating its reproducibility across cachexia models. Since the accumulation of IgG antibodies is a marker of tissue damage and inflammation, we expected this damage to coincide with activation of MuSCs. Indeed, quantification of tdTomato<sup>+</sup> cells by flow cytometry showed a

significant increase in the number of MuSCs in C-26 muscles at cachexia endpoint (Fig. 3D), suggesting their expansion during tumor progression.

To gain further insight into the molecular alterations of MuSCs during cancer cachexia, we utilized our scRNA-seq data to compare transcriptomic differences between control and cachectic MuSCs. Very few cells expressed canonical markers of terminally committed/differentiating myogenic progenitors (*Myod1*, *MyoG*, *Mymk*, *Mymx*) (Supplemental Fig. 2A) which is consistent many other scRNA-seq datasets performed on non-regenerating adult muscles (43, 44). However, pseudobulk differential expression analysis identified 715 differentially expressed genes between control and cachectic MuSCs (Fig. 3E and Supplementary Data 1) and Gene Set Enrichment Analysis (GSEA) revealed coordinated upregulation of pathways associated with acute inflammatory response, striated muscle cell proliferation, and activation of the NF- $\kappa$ B signaling pathway, in cachectic MuSCs (Fig. 3F). Indeed, highly upregulated genes such as *Cfh*, *Slc3914*, *Ier3*, and *Gpx3* (Supplementary Fig. 2B), are known to be induced by muscle damage and inflammation (45-47) and recently identified as markers of 'early activated MuSCs' by scRNA-seq analyses of injured muscles (44, 48). Furthermore, significantly downregulated genes such as *Cdh15*, *Dag1*, and *Ncam1* (Supplementary Fig. 2C), encode membrane proteins that maintain MuSC quiescence by anchoring these cells to their niche between the sarcolemma and basal lamina (5, 49, 50).

Interestingly, GSEA also revealed downregulation of pathways involving skeletal muscle regeneration, cell-cell junction assembly, and myoblast differentiation, in cachectic MuSCs (Fig. 3F). Indeed, downregulated genes such as *Lgals1*, *Cdon*, *Cd81*, *Cd9*, and *Flnc* (Supplementary Fig. 2D), function to mediate cell-cell fusion of committed myogenic progenitors and are critical for terminal differentiation (51-56).

To further resolve the transcriptomic differences in cachectic MuSCs, we isolated the integrated MuSC population and performed subclustering and inferential trajectory analyses. UMAP visualization showed 3 distinct MuSC subclusters (Fig. 3G). Cluster 1 contained a

mixture of control and C-26 MuSCs and was marked by cells with high expression of quiescence markers (*Spry1*, *Hes1*) (Figs. 3H and I)(57, 58). Cluster 2 was predominantly composed of control MuSCs and marked by cells expressing genes that enable myogenic differentiation by promoting cell cycle exit (*Cdkn1a*)(59), myogenic commitment/differentiation (*Myod1*, *Ifrd1*, *Myog*)(60), and myoblast fusion (*Tubb6*)(61) (Figs. 3H and I). Cluster 3 was predominantly composed of C-26 MuSCs and was distinguished by high expression of damage and inflammation related genes such as complement pathway components (*Cfh*, *Serping1*, *C4b*), and NF- $\kappa$ B and Stat3 inflammatory signaling mediators (*Nfkb1*, *Rela*, *Ccl2*, *Cxcl1*, *Slc39a14*, *Tnfrsf1a*, *Stat3*) (Figs. 3H and I). Interestingly, in contrast to the expected downregulation of *Pax7* expression observed in cluster 2, cluster 3 cells maintained high *Pax7* expression levels (Figs. 3H and I), supporting our previous findings that NF- $\kappa$ B activity prevents *Pax7* downregulation in activated myogenic progenitors during cachexia (10). Indeed, pseudotime trajectory analysis revealed a divergence in differentiation trajectory between control and C-26 MuSCs (Fig. 3J). In control MuSCs, trajectory flowed from cells expressing quiescent markers to those expressing canonical commitment/terminal differentiation, while for C-26 MuSCs, trajectories flowed in the opposite direction of commitment/differentiation markers, culminating in the inflammatory cachectic cluster (Fig. 3J). Collectively, these transcriptomic profiles provide additional evidence that cancer cachexia involves chronic muscle damage and inflammation, causing MuSC activation but also impairing their terminal myogenic differentiation.

### **RNA-seq analysis confirms impaired MuSC differentiation during cancer cachexia**

Having confirmed that MuSC dysfunction is a characteristic feature of cancer cachectic muscles, we next sought to determine whether MuSC dysfunction precedes the onset cachexia. Thus, we performed bulk RNA-seq on FACS-isolated tdTomato<sup>+</sup> cells from muscles of control and C-26 tumor-bearing mice 10 days after cancer-cell injection. This time point represents a

period where tumors are established and moderate muscle inflammation occurs (Fig. 3A), but cachexia has not yet occurred. Compared to controls, tdTomato<sup>+</sup> cells from tumor-bearing mice had higher expression of genes associated with early activated but non-committed MuSCs, including *Pax7*, *Myf5*, *Cfh*, *Gpx3*, and *Junb*, and lower expression of genes marking committed and differentiating myogenic progenitors, such as *Myod1*, *Fndc5*, and *Mylk4* (Fig. 4A) (43, 44, 48). The upregulation of early activation genes was confirmed by qRT-PCR on MuSCs isolated from three independent cohorts of C-26 tumor-bearing mice (Fig. 4B). To validate these findings in a separate model of cancer cachexia, we performed similar analysis using our KPCML1 orthotopic PDAC cachexia model (28). As before, TdTomato<sup>+</sup> cells were isolated from hindlimb muscles after allowing tumor establishment, but prior to the onset of muscle wasting. Consistent with the C-26 model, expression of stemness and early activation genes were upregulated in MuSCs of KPCML1 tumor-bearing mice, while no differences in *Myod1* expression were observed (Fig. 4C). These findings demonstrate that in multiple mouse models of cancer cachexia, muscle inflammation and MuSC dysfunction occur in the early stages of tumor progression prior to the onset of cachexia.

After confirming that MuSC dysfunction is a preceding feature of cancer cachexia, we returned to our original question regarding whether aberrant transdifferentiation of non-myogenic mesenchymal progenitors to a myogenic lineage occurs during cachexia. Because scRNA-seq and flow cytometry results disagreed with our previous findings, we critically reassessed our previous study and the literature to seek potential explanations for the discrepancies in results. Interestingly, in most, if not all of the previous studies showing Sca1<sup>+</sup> mesenchymal progenitors could acquire Pax7 and a myogenic fate, experiments were performed using Sca1<sup>+</sup> cells isolated and implanted into regenerating muscles of young animals at ages exhibiting rapid muscle growth (see discussion for details). We realized that in our previous study we used mice at 6-weeks of age which is still within the developmental stage of

muscle growth (62). In contrast, experiments performed in this current study used 12-week-old mice which is considered to be an age when muscle growth is virtually complete (63, 64). Therefore, based on the literature, we speculated that the potential for non-myogenic progenitors to commit to myogenic fate could be restricted to ages of developmental muscle growth. To test this, we performed similar scRNA-seq and flow cytometry experiments shown in Figures 1 and 2, but this time used 6-week-old Pax7-CreER; R26-LSL-tdTomato mice. As before, tamoxifen was administered once tumors were established (12-17 DPI). At endpoint (20 DPI), mononuclear cells were isolated from hindlimb muscles for scRNA-seq and flow cytometry. After quality control filtering, 14,812 cells from control, and 15,274 cells from C-26 muscles, were integrated for clustering analysis. The same 14 major cell types and relative proportions identified in the adult muscles were similarly detected in muscles of young mice (Figs. 5A and B). Furthermore, consistent with the results from adult muscles, expression of myogenic progenitor genes such as *Pax7*, *Myf5*, and *Myod1*, were detected only in the MuSC population (Figs. 5C and D), while the non-myogenic progenitor markers *Ly6a* (*Sca1*) and *Pdgfra* were exclusive to mesenchymal populations such as FAPs, endothelial, and pericytes (Figs. 5C and D). Interestingly, significant *tdTomato* expression was only detected in control MuSCs (Fig. 5D), which may be related to approximately half the number of MuSCs (164 vs 288) being sequenced from C-26 muscles. Indeed, as mentioned previously, the relatively low capture and sequencing depth of lesser abundant cells such as MuSCs, along with the inability to detect genes at the protein level, are inherent limitations of performing unbiased scRNA-seq on multicellular tissues. Therefore, to verify these results we again performed flow cytometry to quantify tdTomato<sup>+</sup> and Sca1<sup>+</sup> cells. C-26 cells were injected into 6 or 20-week-old Pax7-CreER; R26-LSL-tdTomato mice, and tamoxifen was administered 10-15 DPI to label Pax7<sup>+</sup> cells and their progeny with tdTomato. Mononuclear cells were isolated from hindlimb muscles at endpoint for flow cytometry. As expected, the vast majority of tdTomato<sup>+</sup> and Sca1<sup>+</sup> cells separate into distinct cell populations regardless of age or tumor-bearing conditions (Fig. 5E). However, a



minor but evident population of tdTomato<sup>+</sup>; Sca1<sup>+</sup> double positive cells were also observed in the muscles exclusively from young mice (Figs 5G and H). These results indicate that Sca1<sup>+</sup> mesenchymal progenitors can adopt Pax7 expression during cancer cachexia, but this regulation is likely to be restricted to ages of developmental muscle growth.

## DISCUSSION

In our previous study, we found that cachectic muscles from tumor-bearing mice were composed of interstitial cells co-expressing the muscle-specific stem cell marker Pax7, and several non-myogenic progenitor cell markers such as Sca1, Pdgfra, and NG2, suggesting that mesenchymal progenitors were transdifferentiating into myogenic progenitors during cancer cachexia (10). These findings were consistent with what several others had showed that Sca1<sup>+</sup> interstitial cells from mice and human muscle can induce Pax7 expression and adopt a myogenic fate *in vitro* and *in vivo*, following their injection into severely damaged muscles (19, 21, 25, 65-69). However, findings in the current study contest the myogenic conversion of non-myogenic mesenchymal progenitor cells during cancer cachexia. Using scRNA-seq and Pax7-Cre<sup>ER</sup>; tdTomato mice, we show that myogenic and non-myogenic progenitor cells remain as distinct populations in cachectic muscles, running counter to our original model (10).

Does this mean our original model implicating multiple mesenchymal progenitor cells contribute to MuSC dysfunction and muscle wasting in cancer cachexia needs to be revised? Perhaps, given the results from our current lineage tracing and scRNA-seq studies. However, one caveat worth considering is the age at which mice were utilized in our previous studies. After critically evaluating our previous study and existing literature, we realized that in our previous study we used mice that were 6-weeks old (10), which is an age when muscles are still developing (70, 71). Similarly, in most, if not all other studies where Sca1<sup>+</sup> progenitors were

found to induce Pax7 expression and adopt a myogenic fate during muscle regeneration, cellular and transplantation experiments were performed using young animals between 10 days postnatal to 8-weeks old (19, 21, 25, 65-69). These findings are consistent with other published findings from multiple groups simultaneously showing that isolation of Sca1<sup>+</sup>; Cd34<sup>+</sup> interstitial cells from skeletal muscles of young mice ranging from 3-days-old to 4-weeks old undergo myogenesis following their transplantation into severely damaged muscles (25, 65, 67). Other studies have shown that mesoangioblasts or pericytes isolated from 2-week-old dystrophic golden retriever dogs or 3-8 year old Duchenne muscular dystrophy patients, also can commit to a myogenic fate and undergo myogenesis following transplantation into dystrophic muscles (19, 68). Moreover, Sca1<sup>+</sup>; Cd34<sup>+</sup> interstitial cells with Peg3/PW1 expression (PICs) isolated from 10-day old LacZ mice, formed myofibers with  $\beta$ -galactosidase activity after transplantation into freeze-injured muscles (21). These same investigators later reported that these cells were no longer detectable in muscles from 7-week old mice, suggesting that this cell population decreases as muscles reach their full development (66). These results led us to speculate that similar myogenic conversion of Sca1<sup>+</sup> progenitor cells in tumor bearing mice were not seen in our current study due to our utilization of 12-20-week-old mice, which is an age range where developmental muscle growth is no longer occurring (62, 63). Indeed, our flow cytometry results revealing an evident population of Sca1<sup>+</sup>; tdTomato<sup>+</sup> cells in the hindlimb muscles of young mice supports the conclusion that the plasticity observed with non-myogenic progenitors in our previous study (10) was likely a compensatory phenomenon caused by cancer-induced muscle injury and MuSC dysfunction at a time of continued muscle development.

Where our previous model remains consistent with the current scRNA and bulk RNA sequencing analyses, is that tumor burden causes skeletal muscle damage and inflammation, leading to MuSC activation, which in turn impairs differentiation (10, 72). This was evidenced by marked accumulation of IgG in muscle during tumor progression, and scRNA-seq data showing

increased abundance of inflammatory neutrophils and infiltrating monocytes/macrophages in C-26 muscles. This inflammatory phenotype is consistent with recent findings from our laboratory characterizing the regulation of local muscle inflammation in LLC and KPP mice and PDAC patients (73). Muscle damage also causes rapid activation and commitment of MuSCs into a myogenic program (6). Consistent with this notion, our single-cell and bulk transcriptomic analyses showed that cachectic muscles contained activated MuSCs, as evidenced by the upregulation of genes and pathways involved in inflammatory signaling, and the downregulation of genes and pathways that maintain MuSC quiescence. Many of the top differentially expressed genes in cachectic MuSCs, such as of *Gpx3*, *Ier3*, *Junb*, and *Cfh*, were identified as markers of 'early activated MuSCs' in injured muscles (44), while downregulated genes such as *Cdh15* (49, 50) and *Dag1* (74) are important for anchoring MuSCs within their niche which is critical for maintaining quiescence. Subclustering and pseudotime trajectory analyses confirmed that the activated cachectic MuSCs were distinct from activated control MuSCs and marked by greater expression of genes related to tissue damage and inflammation (*Cfh*, *Serping1*, *C4b*, *Ccl2*, *Cxcl1*, *Slc39a14*, *Tnfrsf1a*, *Nfkb1*), and lesser expression of genes necessary for myogenic commitment and differentiation (*Ifrd1*, *Cdkn1a*, *Lgals1*). Similar findings have recently been reported using the LLC cachexia model (75). Interestingly, in addition to its role in promoting cell cycle exit, upregulation of *Cdkn1a* (p21) during C2C12 myoblast differentiation was shown to be important for preventing apoptosis (76). Thus, the lower expression of *Cdkn1a* in activated cachectic MuSCs might contribute to their clearance following their impaired fusion to damage myofibers, as we previously described (77).

With regards to limitations of our study, one potential limitation is that our tamoxifen dosing to label Pax7<sup>+</sup> cells was done within a specific timeframe of tumor progression (12-17 DPI) which overlaps with muscle inflammation and myofiber atrophy (11) but still prior to the severity of body weight loss used to classify cachexia (~21 DPI). It's possible that the acquisition of Pax7

expression by Sca1<sup>+</sup> progenitors does not occur until advanced stages when cachexia has developed, in which case, our tamoxifen dosing period would have been too early to label Sca1<sup>+</sup>; Pax7<sup>+</sup> cells with tdTomato. Another potential limitation is our decision to forego a pre-sequencing purification step to isolate only tdTomato<sup>+</sup> cells for downstream scRNA-seq which limited resolution of tdTomato<sup>+</sup> cell populations. We made this because of the additional stress that FACS-isolation places on cells which influences their transcriptional profiles (78) and potentially prevent detection of novel cell types in cachectic muscles. However, this resulted in our analyses being dominated by the more abundant mononuclear cell types of the muscle microenvironment such as FAPs and endothelial cells and only a relatively small number of MuSCs being sequenced. This limits both total sequencing depth of the myogenic progenitors and capture of cells within the highly dynamic terminal commitment and fusion stages of myogenesis (43).

Taken together, the current results are supportive of our original model, but the new data generated warrants a revision of the model. As shown in the graphical abstract, during the induction and progression of cancer in the adult setting, as would occur in PDAC, factors released from the tumor cause skeletal muscle injury and local inflammation. This muscle damage leads to Pax7<sup>+</sup> MuSCs to break from quiescence in their niche and expand. This chronic inflammatory muscle microenvironment causes sustained NF- $\kappa$ B activation and Pax7 expression in the activated myogenic progenitors, resulting in impaired terminal differentiation, unrepaired myofibers, and muscle atrophy. However, contrary to our previous model, the transdifferentiation of Sca1<sup>+</sup> mesenchymal progenitors is likely a developmental age dependent phenomenon and therefore does not contribute to the pool of dysfunctional Pax7<sup>+</sup> myogenic progenitors in adult muscle. Moving forward, additional studies are likely to continue refining specific aspects of this model by which skeletal muscle damage, NF- $\kappa$ B activation, and MuSC dysfunction, contribute to muscle wasting during cancer by impairing myogenesis.

## **SUPPLEMENTAL MATERIAL**

**Supplemental Figs. 1-3.** 10.6084/m9.figshare.33165254

**Supplemental Data 1.** 10.6084/m9.figshare.33166628

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## **CONTRIBUTIONS**

S.G.M., M.C.O., T.A.Z., and D.C.G. conceptualized, designed the experiments. and wrote the manuscript. S.G.M, D.J.W, A.U., E.F., J.O, were involved in the methodology. S.G.M. and D.C.G. planned and outlined the manuscript, and wrote the original draft, while editing was performed by all the authors. S.G.M., C.A.Q., S.S., and S.B. performed the bioinformatics and statistical analysis.

## **FIGURE LEGENDS**

**Figure 1. ScRNA-seq of skeletal muscle reveals distinct separation of myogenic and non-myogenic progenitors during cancer cachexia.**

**(A)** Schematic representation of experimental design. **(B)** Change in body weight of control or C-26 tumor-bearing mice at time of sacrifice (21-days post injection). N=2/group, 12-wks-old. **(C)** Uniform Manifold Approximation Projection (UMAP) of mononuclear cells isolated from

hindlimb skeletal muscles of control (16,427) and C-26 (11,528) muscles. **(D)** Annotated UMAP of the identified clusters and cell types. **(E)** Dot plot showing gene expression of marker genes used to classify cell types. **(F)** UMAPs showing myogenic (*Pax7*, *tdTomato*) and **(G)** non-myogenic (*Ly6a*, *Pdgfra*) progenitor gene expression in mononuclear cells of control vs C-26 tumor-bearing muscles. **(H)** Violin plots showing myogenic (*Pax7*, *tdTomato*) and non-myogenic (*Ly6a*, *Pdgfra*) progenitor gene expression across cell types in control and C-26 muscle mononuclear cells.

**Figure 2. Confirmation by flow cytometry that myogenic and non-myogenic progenitors remain distinct populations in skeletal muscle during cancer cachexia.**

**(A)** Representative flow cytometry scatter plots showing Sca1 and tdTomato expressing cells on mononuclear cells isolated from muscles of control or C-26 tumor-bearing mice. N=3/condition, 13-wks-old. **(B)** Quantification of the percentage of cells expressing Sca1, tdTomato, or Sca1 and tdTomato. **(C)** mRNA expression of myogenic and non-myogenic progenitor genes in FACS isolated cells from control or C-26 muscles. N=pooled cells from 3 independent samples, 12-wks-old. **(D)** Sca1 protein expression in FACS-isolated tdTomato<sup>+</sup>/tdTomato<sup>-</sup> cell populations, 14-wks-old.

**Figure 3. Tissue inflammation and MuSC activation in cancer cachectic muscle.**

**(A and B)** Tibialis anterior muscles from C-26 or LLC tumor-bearing mice were harvested at 0, 10, 14, 18, and 21 days-post cancer cell injection. Extracted proteins were separated by SDS-PAGE and incubated with anti-mouse IgG-HRP conjugated antibody. 16-wks-old **(C)** Tibialis anterior muscles proteins from control or cachectic KPCML tumor bearing mice incubated with mouse IgG-HRP conjugated antibody. 16-wks-old. **(D)** Flow cytometry-based quantification of tdTomato<sup>+</sup> cells per mg of muscle tissue in control versus C-26 tumor bearing mice. N=3-

4/group, 18-wks-old. Two-tailed unpaired  $t$  test \*\*\*  $p < 0.001$ . **(E)** Volcano plot showing differentially expressed genes between control vs C-26 MuSCs identified in scRNA-seq. dataset. **(F)** Significantly upregulated and downregulated biological pathways identified by Gene Set Enrichment Analysis (GSEA) between control vs. C-26 MuSCs. **(G)** UMAP plot of MuSC subclusters. **(H)** UMAP plots of MuSC subclusters colored by indicated genes. **(I)** Dot plots of subcluster gene markers. **(J)** UMAP plots showing pseudotime trajectory analysis of control vs C-26 MuSCs. Trajectory begins in quiescent cluster (dark blue).

**Figure 4. Dysfunctional MuSC differentiation precedes the onset of muscle wasting in multiple cancer cachexia models. (A)** Heat map generated from bulk RNA-seq analysis of FACS-purified tdTomato<sup>+</sup> cells from muscles of control or C-26 tumor bearing mice at day 10 post-injection. N=pooled cells from hindlimb muscles of 3 mice/group, 12-wks-old. **(B)** Expression of select genes from (A) were confirmed by qRT-PCR. Data plotted as mean  $\pm$  SD; N=pooled cells from 3 mice/replicate, 12-14-wks-old. Two-tailed unpaired  $t$  test, \* $p < 0.05$ , \*\* $p < 0.01$ . **(C)** qRT-PCR was performed for similar genes as in (B) using mRNA from FACS-isolated tdTomato<sup>+</sup> cells from muscles of control or KPCML1 tumor bearing mice. Data plotted as mean  $\pm$  SD; N=pooled cells from 3 mice/replicate, 20-wks-old. Two-tailed unpaired  $t$  test, \* $p < 0.05$ .

**Figure 5 Assessing progenitor cell fates in developing muscles during cachexia.**

**(A)** Integrated UMAP of mononuclear cells isolated from control (14,812) and C-26 (15,274) muscles. N=2/group, 6-wks-old. **(B)** Bar plots showing total number and relative proportions of cell types in control and C-26 muscles. **(C)** Dot plot showing gene markers for individual cell types. **(D)** UMAPs showing gene expression of myogenic (*Pax7*, *tdTomato*) and non-myogenic (*Ly6a*, *Pdgfra*) progenitor cells in control vs C-26 muscles. **(E)** Representative flow cytometry

plots showing Sca1<sup>+</sup>, tdTomato<sup>+</sup>, and Sca1<sup>+</sup>tdTomato<sup>+</sup> cells from hindlimb muscles of mice injected with either PBS (control) or C-26 cancer cells (C-26) at either 20-weeks (Adult) or 6-weeks (Young) of age. **(F and G)** Quantification of cell populations expressed relative to total live cells measured in adult (B) or young (C) mice. **(H)** Quantification of Sca1<sup>+</sup>; tdTomato<sup>+</sup> cells relative to total tdTomato<sup>+</sup> cells. Full gating strategies are shown in Supplemental Figure 3.



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FIGURE 1

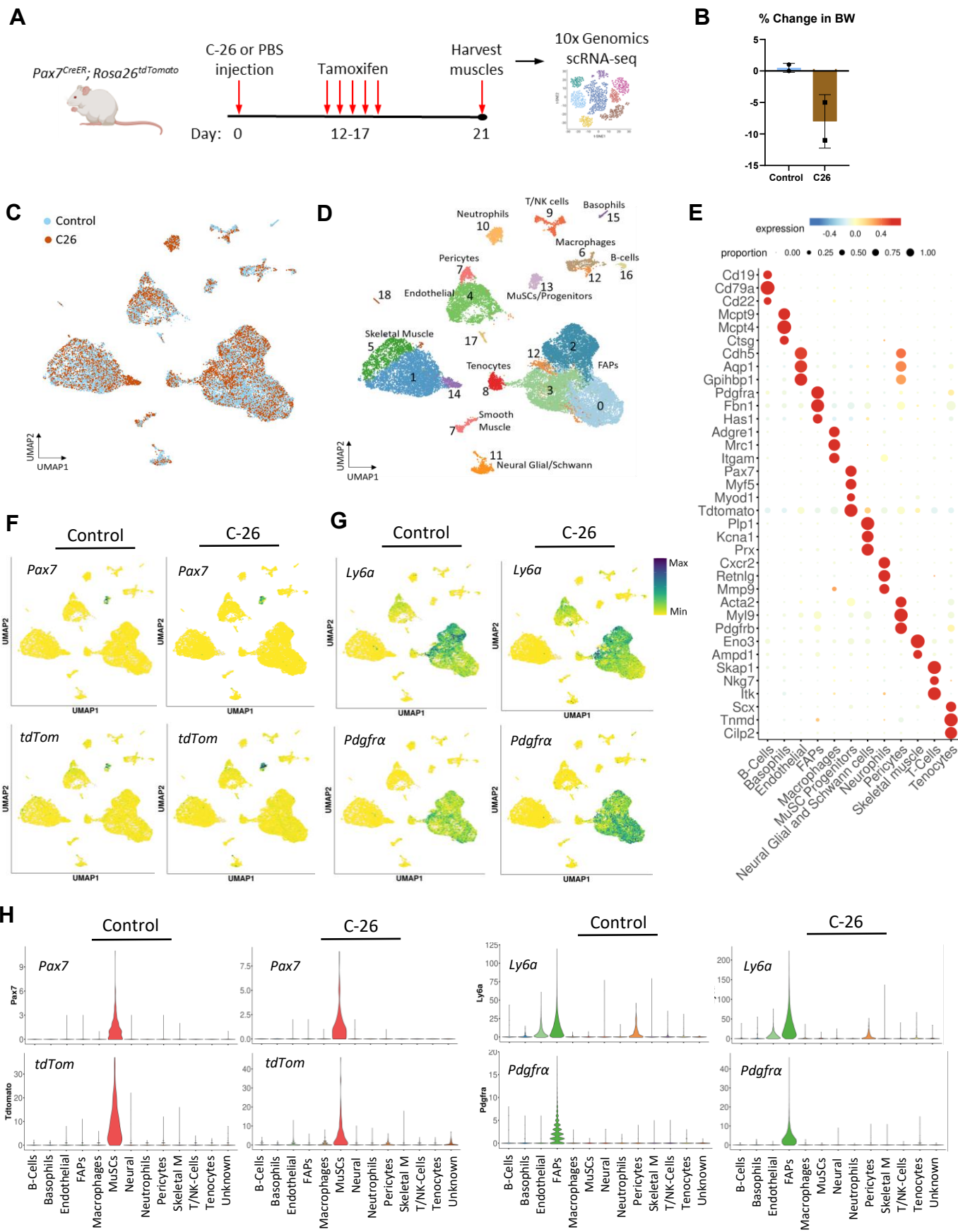


FIGURE 2

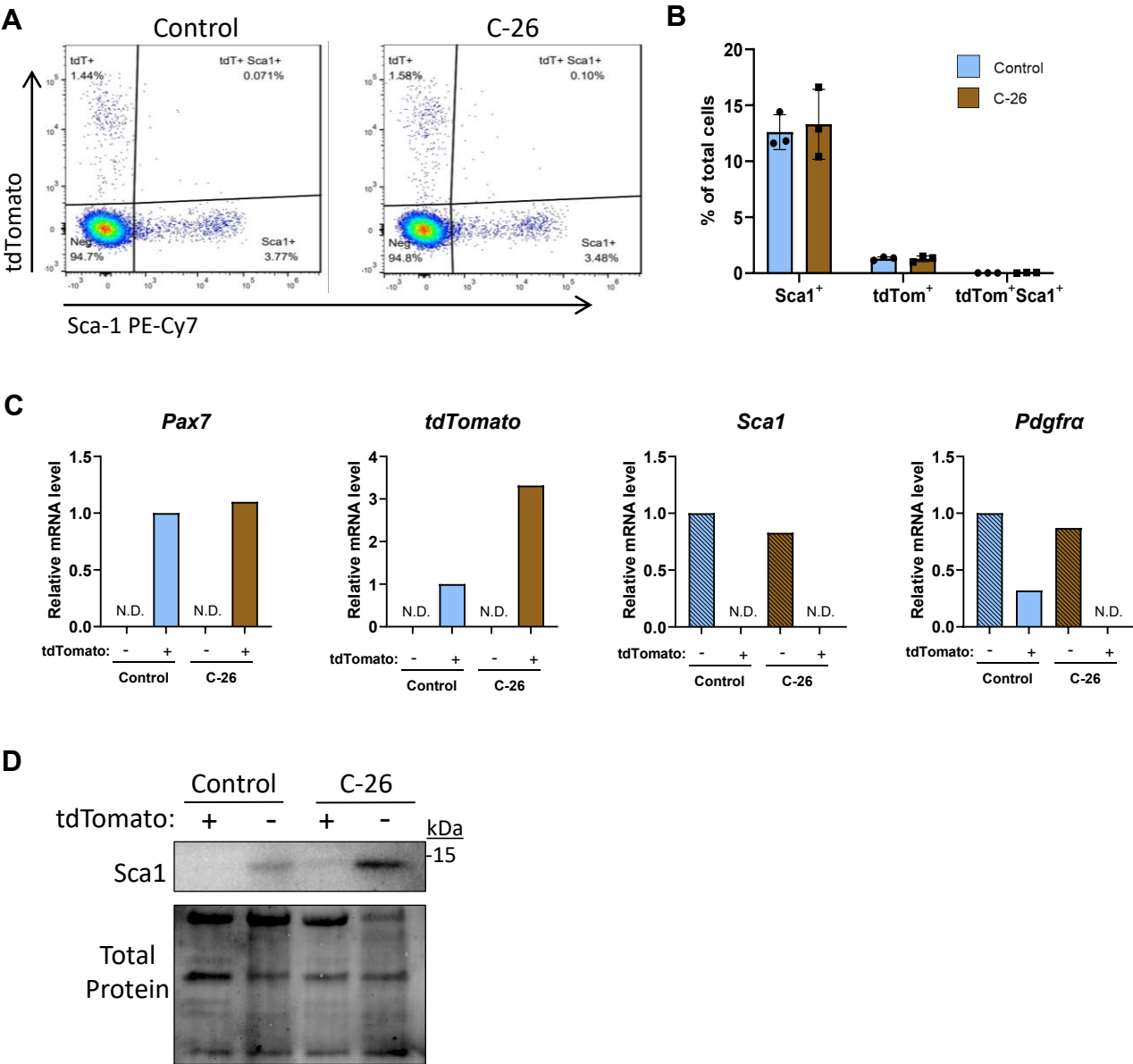


FIGURE 3

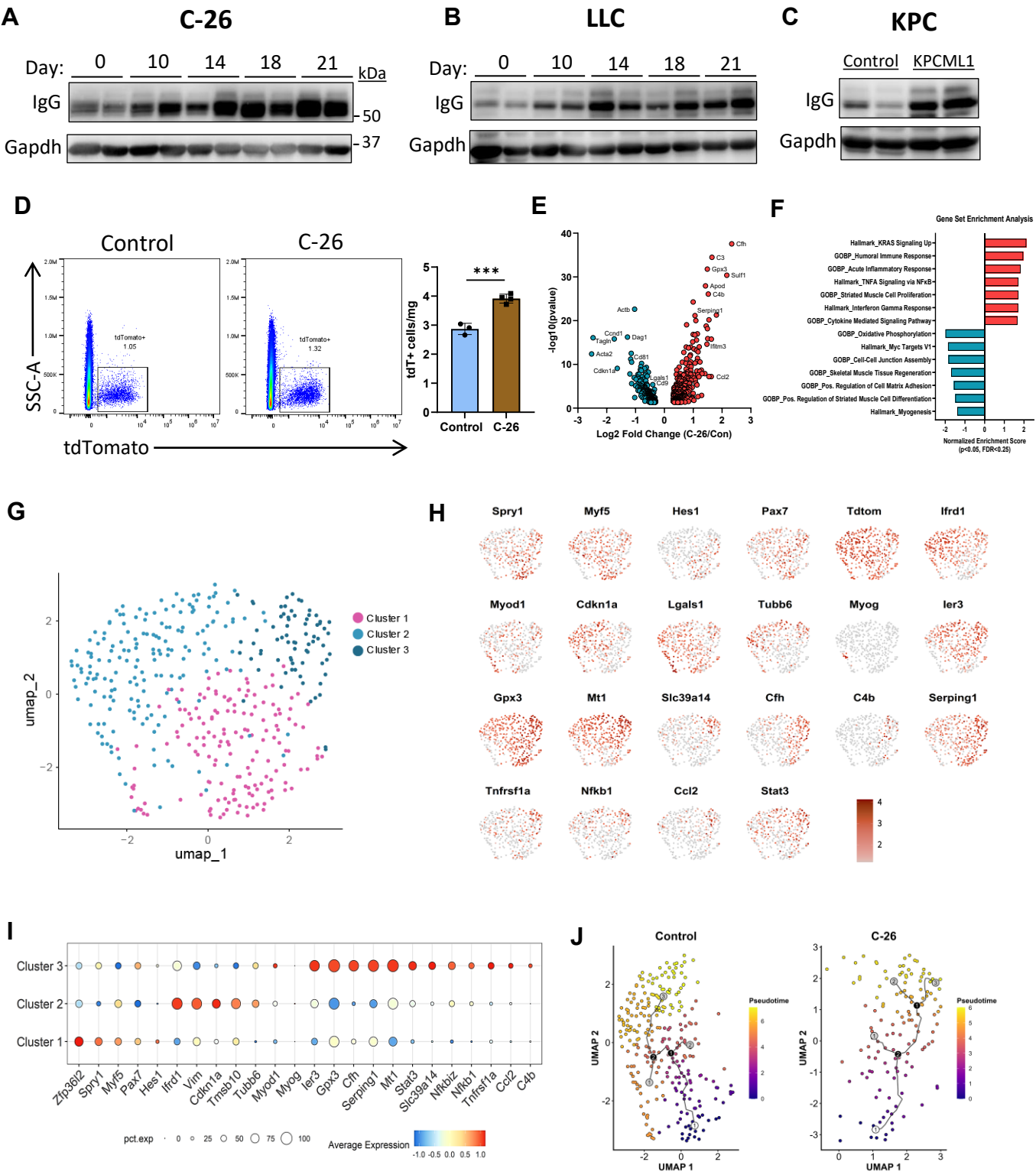




FIGURE 4

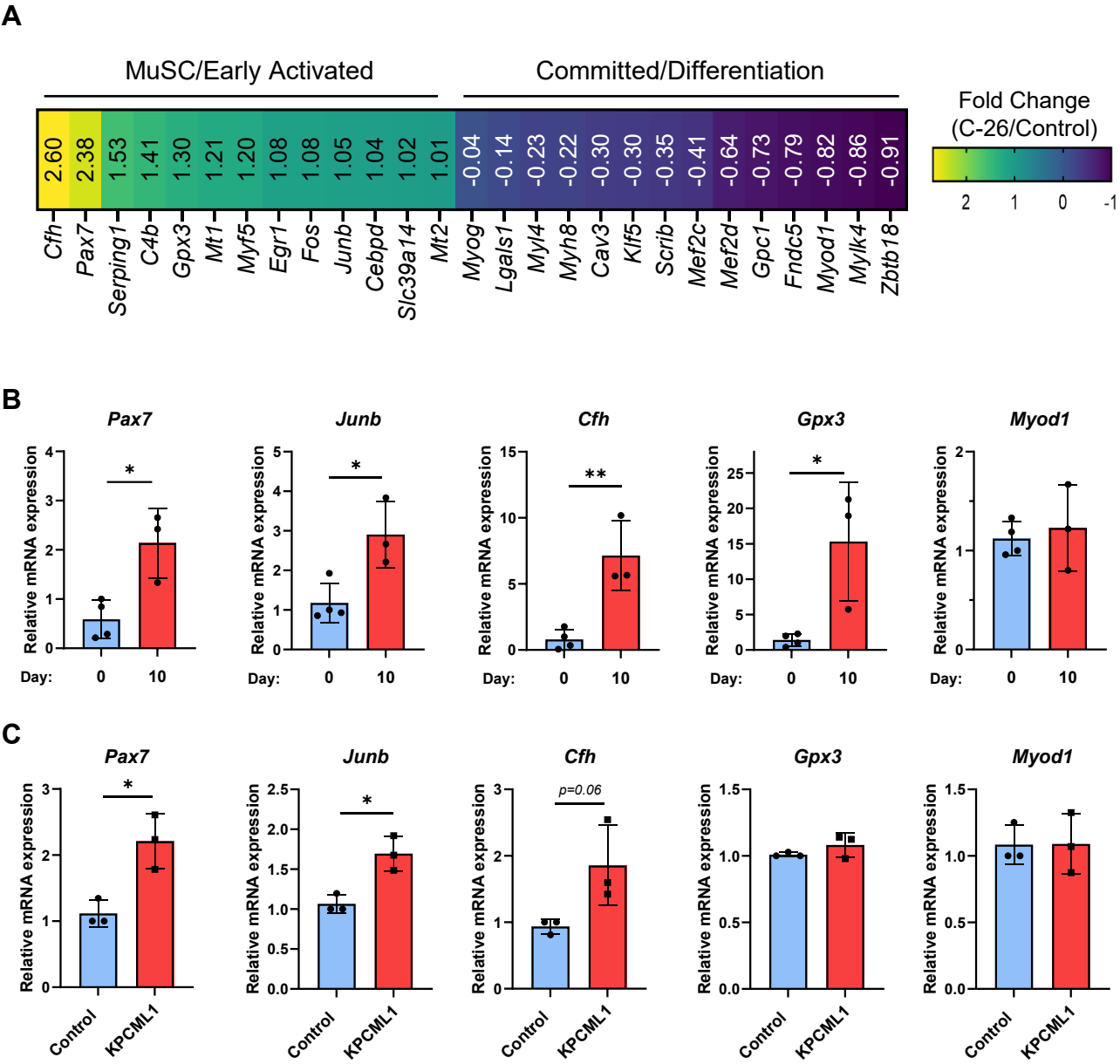
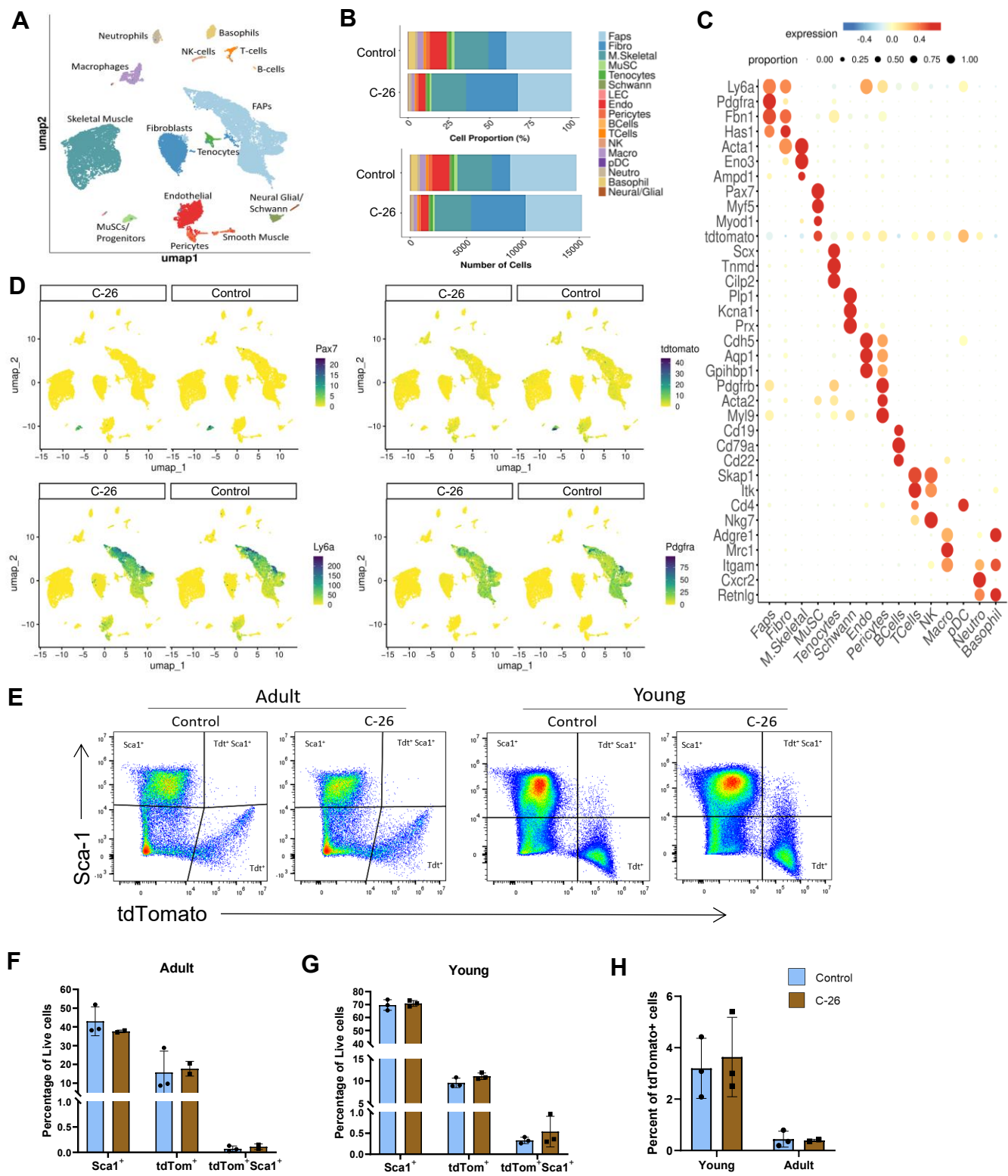


FIGURE 5



## A revised model for the role of muscle resident progenitor cells in cancer cachexia

